



Excel Data Analysis

Generating Insights Through Mid-Year Sales Data

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Business Case

Companies aim to track their sales performance and optimize their sales strategies to achieve the targets set for the end of the year. They plan to analyze sales data from the past 6 months by examining various metrics that measure the performance of both sales and sales agents. This analysis will provide insights for decision-making, helping to guide sales strategies for the next 6 months.

Objective

- Gain insight into sales performance based on key metrics
- Analyze sales performance trends over time to understand areas of improvement
- Assess sales performance based on various variables
- Evaluate sales agents performance through various assessment metrics
- Provide data-driven recommendations to guide sales improvement strategies and enhance the performance of sales agents.

Dataset Overview

Consists of 3 data sheets, namely lead data, sales data, and sales agents target data. Contains a total of 797 data

Raw dataset: <https://docs.google.com/spreadsheets/d/1j-akwvUQuQeojbrc9uuxFc85j8jGmk52/edit?usp=sharing&ouid=111194686740125574873&rt=pof=true&sd=true>

Leads Data

S.No.	Customer Name	Sector	City	State	Postal Code	Region	Sales person assigned	Lead date
1	Aaron Bergman	Retail	Henderson	Kentucky	42420	South	Michael	28/01/2022
2	Justin Ritter	Retail	Los Angeles	California	90036	West	Andy	05/06/2022
3	Craig Reiter	Technology	Fort Lauderdale	Florida	33311	South	Angela	16/04/2022
4	Katherine Murray	Financial	Los Angeles	California	90032	West	Meredith	22/05/2022
5	Rick Hansen	Healthcare	Concord	North Carolina	28027	South	Toby	11/06/2022
6	Jim Mitchum	Technology	Seattle	Washington	98103	West	Jim	03/01/2022

- **Customer Name:** name of leads
- **Sector:** leads business sector
- **City:** leads city
- **State:** leads state
- **Postal Code:** leads address postal code
- **Region:** leads region
- **Sales Person Assigned:** sales agent who make contact with leads
- **Lead date:** the date the leads was first reached

Sales Data

S.No.	Customer Name	Sales person assigned	Category	Order date	Sales
1	Craig Reiter	Angela	Envelopes	14/05/2022	\$ 576,65
2	Katherine Murray	Meredith	Letterhead	13/06/2022	\$ 1.314,33
3	Rick Hansen	Toby	Letterhead	02/07/2022	\$ 494,49
4	Anthony Jacobs	Ryan	Copy Paper	11/06/2022	\$ 409,00
5	Peter Fuller	Dwight	Envelopes	21/06/2022	\$ 422,07
6	Ben Peterman	Pam	Letterhead	28/02/2022	\$ 537,68

- **Customer Name:** name of customers
- **Sales Person Assigned:** sales agent who responsible for the sales deals
- **Category:** product category purchased by customers
- **Order date:** purchase date
- **Sales:** sales value

Sales Person Target Data

Sales person Target	
Andy	\$ 21.300
Angela	\$ 18.100
Dwight	\$ 17.800
Jim	\$ 22.800
Kelly	\$ 20.600

- **Sales Person:** name of sales agent
- **Target:** sales agent's annual sales target

Sales Performance Metrics

Metrics used to track and assess sales performance

Sales Metrics

Key Metrics:

- **Sales Volume**
 - Track the progress of current sales volume against year-end targets
- **Avg Growth Rate**
 - Assess the effectiveness of the implemented sales strategy
- **Conversion Rate**
 - Assess the effectiveness segmentation strategies and sales approaches
- **Avg Sales Cycle Length**
 - Assess the efficiency of the sales process

Sales Agent Metrics

Sales Agent Performance Metrics:

- **Sales Volume per Agent**
 - Gauge individual performance, identify top performers, and implement targeted coaching or incentives
- **Winrate**
 - Measure individual effectiveness in converting leads into sales
- **Quota Attainment**
 - Gauge individual success in exceeding assigned sales targets

Other Metrics:

- **Sales by City**
 - Assess regional market performance
- **Sales by Product Category**
 - Assess product line performance
- **Average Order Value per Product Category**
 - Gain insights into purchasing behavior and product preferences
- **Sales by Customer Sector**
 - Identify opportunities based on sector-specific insights
- **Sales by Customer**
 - Identify high-value customers

Data Processing

Since the existing data is already clean, free from formatting errors, inaccuracies, duplication, etc, it can be processed immediately.

S.No	Customer Name	Sector	City	State	Postal Code	Region	Sales person assigned	Lead date	Category	Order date	Sales	Converted	Conversion Day
1	Aaron Bergman	Retail	Henderson	Kentucky	42420	South	Michael	28/01/2022	Not Converted	Not Converted	Not Converted	No	Not Converted
2	Justin Ritter	Retail	Los Angeles	California	90036	West	Andy	05/06/2022	Not Converted	Not Converted	Not Converted	No	Not Converted
3	Craig Reiter	Technology	Fort Lauderdale	Florida	33311	South	Angela	16/04/2022	Envelopes	14/05/2022	\$ 576,65	Yes	28
4	Katherine Murray	Financial	Los Angeles	California	90032	West	Meredith	22/05/2022	Letterhead	13/06/2022	\$ 1,314,33	Yes	22
5	Rick Hansen	Healthcare	Concord	North Carolina	28027	South	Toby	11/06/2022	Letterhead	02/07/2022	\$ 494,49	Yes	21
6	Jim Mitchum	Technology	Seattle	Washington	98103	West	Jim	03/01/2022	Not Converted	Not Converted	Not Converted	No	Not Converted

Feature Engineering

- Extract the category data, order date, and sales value from the sales data
- Group the leads into those that were successfully converted and those that were not
- Aggregate the lead date and order date data to calculate the sales cycle length
- Consolidate all required data into a leads data sheet for centralized data source

Function

- VLOOKUP
- INDEX MATCH
- IF
- IFERROR

Pivot Table

Key Metrics

Total Revenue (mid-year) vs Yearly Target

Sales Volume	Annual Target	Progress	Remaining
\$ 263.683	\$ 500.000	53%	47%
\$263.683			

Total Lead & Conversion Rate

Converted		Yes	
Total Lead	Total Customer	Conversion Rate	Remaining
796	320	40%	60%
796			

Avg Sales/month & Avg Growth Rate/month

Month	Total Sales	Growth Rate
Jan	\$27.423	
Feb	\$49.029	79%
Mar	\$54.910	12%
Apr	\$26.655	-51,5%
May	\$59.871	125%
Jun	\$45.796	-24%
Avg Sales/Month	\$43.947	
Avg Growth/Month		28%
	remaining	72%

Avg Sales Cycle Length

Avg Sales Cycle Length
16
16

Pivot Table

Sales Performance

Sales vs Lead Trend/month

Month	Total Sales	Total Leads
Jan	\$27.423	134
Feb	\$49.029	120
Mar	\$54.910	123
Apr	\$26.655	126
May	\$59.871	157
Jun	\$45.796	136
Grand Total	\$263.683	796

Sales by City

City	Total Sales
Burlington	\$4.076
Columbia	\$4.211
Dallas	\$5.422
Chicago	\$7.207
Louisville	\$9.811
Seattle	\$13.437
Los Angeles	\$15.110
Philadelphia	\$16.794
New York City	\$21.022
San Francisco	\$22.951
Grand Total	\$120.041

Sales by Cust Sector

Sector	Total Sales
Wholesale	\$25.641
Retail	\$31.554
Manufacturing	\$32.004
Healthcare	\$32.412
Service	\$35.578
Financial	\$50.660
Technology	\$55.834
Grand Total	\$263.683

Total Sales & Avg Order Value by Product

Product	Total Sales	Avg Order Value
Copy Paper	\$131.746	\$878
Envelopes	\$72.995	\$737
Letterhead	\$58.942	\$830
Grand Total	\$263.683	\$824

Pivot Table

Sales Agents Performance

Sales Agents Winrate

Win Rate		
Agent	No	Yes
Andy	54,9%	45,1%
Angela	61,8%	38,2%
Dwight	59,5%	40,5%
Jim	70,5%	29,5%
Kelly	57,1%	42,9%
Meredith	50,7%	49,3%
Michael	53,7%	46,3%
Pam	59,7%	40,3%
Ryan	62,9%	37,1%
Toby	62,8%	37,2%

Sales Agents Total Sales vs Target & Winrate

Agent	Total Sales	Annual Target	Quota Attainment	Winrate
Andy	\$27.261	\$ 21.300	128%	45%
Angela	\$18.801	\$ 18.100	104%	38%
Dwight	\$27.188	\$ 17.800	153%	40%
Jim	\$19.597	\$ 22.800	86%	29%
Kelly	\$24.932	\$ 20.600	121%	43%
Meredith	\$25.452	\$ 23.700	107%	49%
Michael	\$34.970	\$ 25.400	138%	46%
Pam	\$23.072	\$ 23.300	99%	40%
Ryan	\$28.663	\$ 20.400	141%	37%
Toby	\$33.746	\$ 29.500	114%	37%
Grand Total	\$263.683			

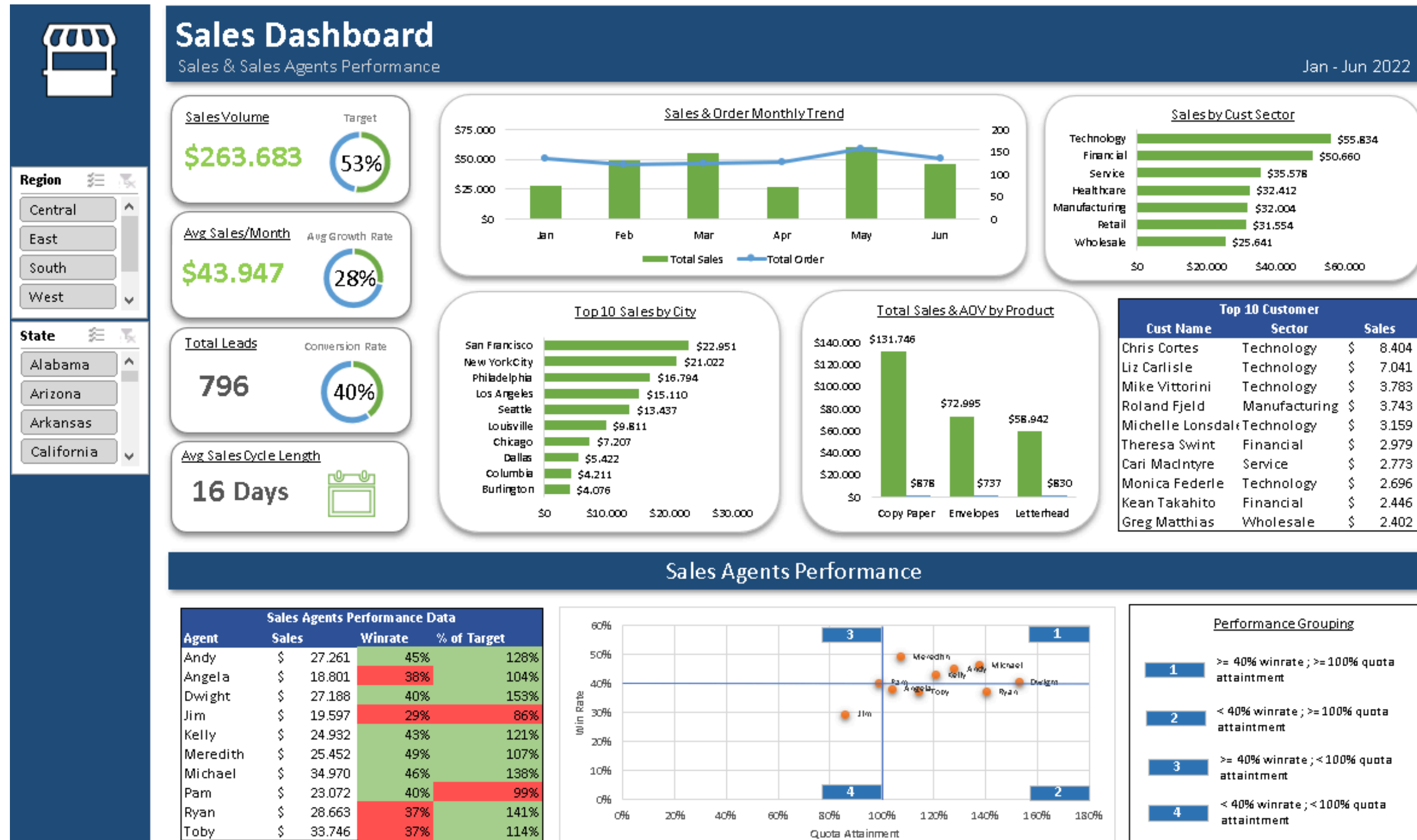
Pivot Table

Customers Data

Customer Total Sales

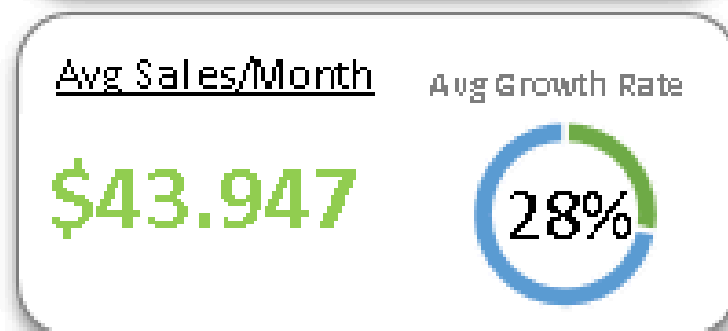
Customer	Total Sales	Sector	Total Sales non pivot
Chris Cortes	\$8.404	Technology	\$ 8.404
Liz Carlisle	\$7.041	Technology	\$ 7.041
Mike Vittorini	\$3.783	Technology	\$ 3.783
Roland Fjeld	\$3.743	Manufacturing	\$ 3.743
Michelle Lonsdal	\$3.159	Technology	\$ 3.159
Theresa Swint	\$2.979	Financial	\$ 2.979
Cari MacIntyre	\$2.773	Service	\$ 2.773
Monica Federle	\$2.696	Technology	\$ 2.696
Kean Takahito	\$2.446	Financial	\$ 2.446
Greg Matthias	\$2.402	Wholesale	\$ 2.402
Grand Total	\$39.428		

Sales Performance Dashboard



Insights & Recommendation

Sales Performance Key Metrics



Insights:

- Sales Volume have surpassed 50% of the year-end goal within 6 months, indicates strong sales performance, aligning with estimates within the expected time frame
- The 28% average growth rate over 6 months reflect the effectiveness of sales and marketing strategies implemented during this period
- A conversion rate of 40% indicates a strong effectiveness in converting leads into sales, signaling efficient sales processes and potentially high-quality leads
- An avg sales cycle length of 16 days indicates prompt response times, effective communication, and streamlined procedures, contributing to quicker conversions and revenue generation

Recommendation:

To capitalize on these strengths, the company should focus on scaling successful strategies, optimizing lead management processes, streamlining sales procedures further, adjusting targets in alignment with exceeded sales volumes, and enhancing overall customer experience to drive sustained growth and success.

Insights & Recommendation

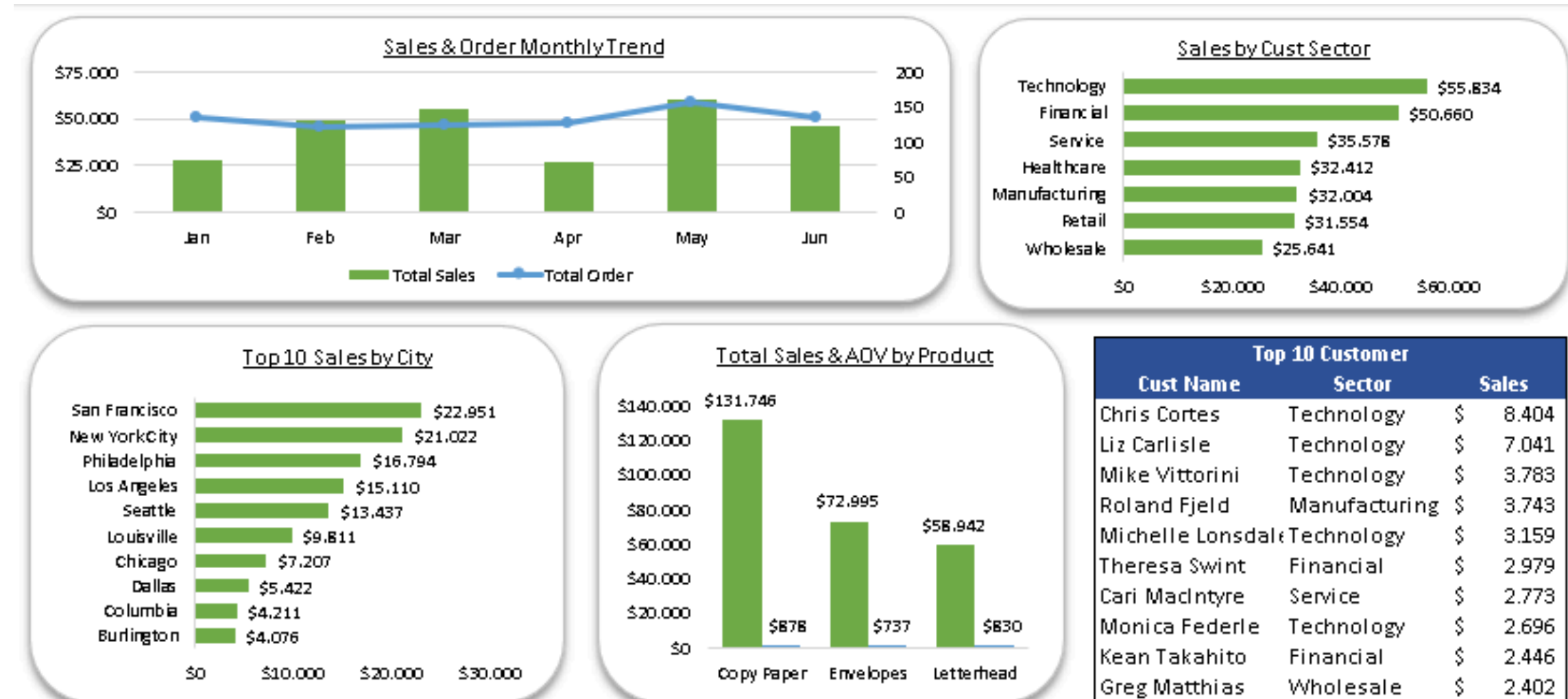
Sales Performance Metrics

Insights:

- Sales and leads trends fluctuate throughout the last 6 months, with no apparent correlation between them
- In April, there was a notable decrease in sales, despite an upward trend in leads compared to the previous month
- The majority of cities that generate the most sales come from the east and west regions
- The most frequently purchased product category is copy paper, with the average order value remaining consistent across all product categories
- High-value customers primarily originate from the technology and financial sectors.

Recommendation:

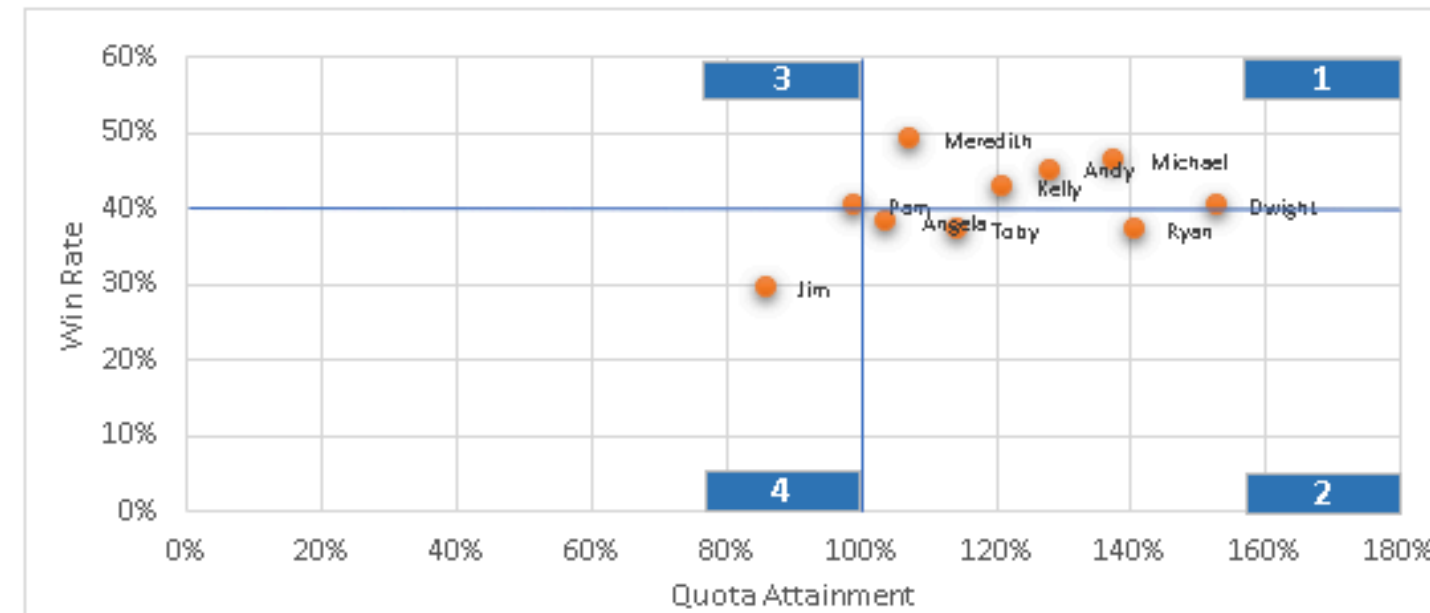
With sales trends and leads fluctuating, and no visible correlation between the two variables, companies need to conduct a deeper analysis and explore other factors that may influence increased sales. The significant decline in sales in April necessitates an evaluation of the company's performance during that month. By leveraging comprehensive sales data across various variables, companies can tailor their sales strategy towards cities, products, and sectors with high sales value.



Insights & Recommendation

Sales Agents Performance Metrics

Sales Agents Performance Data				
Agent	Sales	Winrate	% of Target	
Andy	\$ 27.261	45%	128%	
Angela	\$ 18.801	38%	104%	
Dwight	\$ 27.188	40%	153%	
Jim	\$ 19.597	29%	86%	
Kelly	\$ 24.932	43%	121%	
Meredith	\$ 25.452	49%	107%	
Michael	\$ 34.970	46%	138%	
Pam	\$ 23.072	40%	99%	
Ryan	\$ 28.663	37%	141%	
Toby	\$ 33.746	37%	114%	



Performance Grouping	
1	$\geq 40\%$ winrate ; $\geq 100\%$ quota attainment
2	$< 40\%$ winrate ; $\geq 100\%$ quota attainment
3	$\geq 40\%$ winrate ; $< 100\%$ quota attainment
4	$< 40\%$ winrate ; $< 100\%$ quota attainment

To assess the performance of sales agents, they are grouped based on win rate and quota attainment, as detailed in the performance grouping box

Insights:

- The majority of sales agents are in groups 1 and 2, indicating that most have met the specified targets and have a good win rate
- There are still sales agents who have not met the win rate standards despite having met their targets
- There are two sales agents who have not met the specified targets, and one of them also has a fairly low win rate

Recommendation:

Leverage top performers in groups 1 and 2 to mentor others and share best practices. Support group 3 with additional resources and training to help them achieve their quotas. Develop group 4 through targeted training and one-on-one coaching, setting clear, achievable goals. Introduce performance-based incentives to motivate all sales agents and optimize sales processes using insights from high performers.

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Thank You !

~ Glad to receive your input
and feedback ~

Full Data

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Linkedin

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